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Network Engineering

IPv4 & IPv6

Task 1 - Ports and Protocols

1. Using topology from previous lab, replaced Worker VPCS with Worker Ubuntu Cloud Guest:

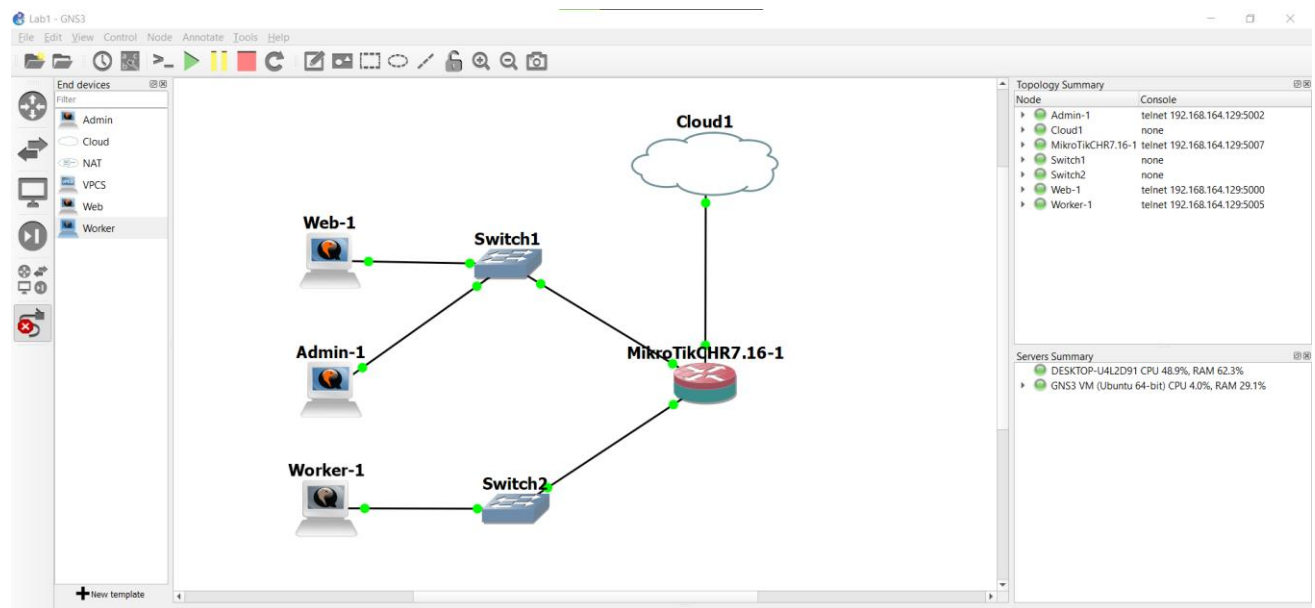


Figure 1 Topology for Task 1

Checking open ports on Web and Admin machines:

```
ubuntu@ubuntu-cloud:~$ sudo netstat -at
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:domain        0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:http            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh              0.0.0.0:*               LISTEN
tcp6       0      0 [::]:http               [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                 [::]:*                  LISTEN
ubuntu@ubuntu-cloud:~$ sudo lsof -i TCP:80
COMMAND PID  USER  FD  TYPE DEVICE SIZE/OFF NODE NAME
nginx   619   root   6u  IPv4 17919      0t0  TCP *:http (LISTEN)
nginx   619   root   7u  IPv6 17920      0t0  TCP *:http (LISTEN)
nginx   620 www-data 6u  IPv4 17919      0t0  TCP *:http (LISTEN)
nginx   620 www-data 7u  IPv6 17920      0t0  TCP *:http (LISTEN)
ubuntu@ubuntu-cloud:~$
```

```
ubuntu@ubuntu-cloud:~$ sudo ss -tulnp | grep 80
tcp  LISTEN  0      511      0.0.0.0:80      0.0.0.0:*      users:((("nginx",pid=588,fd=6),("nginx",pid=587,fd=6)))
tcp  LISTEN  0      511      [::]:80        [::]:*        users:((("nginx",pid=588,fd=7),("nginx",pid=587,fd=7)))
ubuntu@ubuntu-cloud:~$
```

Figure 2 Checking open ports on Web with lsof, netstat and ss

```
ubuntu@ubuntu-cloud:~$ sudo netstat -at
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:domain        0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh              0.0.0.0:*               LISTEN
^C
ubuntu@ubuntu-cloud:~$ sudo lsof -i :22
COMMAND PID  USER  FD  TYPE DEVICE SIZE/OFF NODE NAME
sshd    587   root   3u  IPv4 17817      0t0  TCP *:ssh (LISTEN)
sshd    587   root   4u  IPv6 17828      0t0  TCP *:ssh (LISTEN)
ubuntu@ubuntu-cloud:~$
```

```
ubuntu@ubuntu-cloud:~$ sudo ss -tulnp | grep 22
tcp  LISTEN  0      128      0.0.0.0:22      0.0.0.0:*      users:((("sshd",pid=580,fd=3)))
tcp  LISTEN  0      128      [::]:22        [::]:*        users:((("sshd",pid=580,fd=4)))
ubuntu@ubuntu-cloud:~$
```

Figure 3 Checking open ports on Admin with lsof, netstat and ss

NAT options on gateway:

```
[admin@MikroTik] > ip firewall nat print
Flags: X - disabled, I - invalid; D - dynamic
0 chain=srcnat action=masquerade out-interface=ether3

1 chain=dstnat action=dst-nat to-addresses=192.168.10.2 to-ports=22 protocol=tcp in-interface=ether
3 dst-port=2222

2 chain=dstnat action=dst-nat to-addresses=192.168.10.3 to-ports=22 protocol=tcp in-interface=ether
3 dst-port=2223

3 chain=dstnat action=dst-nat to-addresses=192.168.10.2 to-ports=80 protocol=tcp in-interface=ether
3 dst-port=8080
[admin@MikroTik] > []
```

Figure 4 NAT options on gateway

2. Scanning gateway from Ubuntu VM which running gns3 server:

```
salekh@salekhmamadaliev25sne01:~$ nmap -T4 -p- -n 192.168.49.2
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-11 08:51 UTC
Nmap scan report for 192.168.49.2
Host is up (0.0052s latency).
Not shown: 65524 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
80/tcp    open  http
2000/tcp  open  cisco-sccp
2222/tcp  open  EtherNetIP-1
2223/tcp  open  rockwell-csp2
8080/tcp  open  http-proxy
8291/tcp  open  unknown
8728/tcp  open  unknown
8729/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 27.77 seconds
```

Figure 5 Gateway scan result with enabled services

Known open ports are 21-ftp, 22-ssh,23-telnet,80-http.

3. Disabled unnecessary services on gateway:

```
[admin@MikroTik] > ip service print
Flags: X - DISABLED, I - INVALID
Columns: NAME, PORT, CERTIFICATE, VRF, MAX-SESSIONS
#  NAME      PORT  CERTIFICATE  VRF  MAX-SESSIONS
0 X telnet    23          main      20
1 X ftp       21          main      20
2 X www       80          main      20
3 X ssh       22          main      20
4 X www-ssl   443  none        main      20
5 X api       8728         main      20
6 X winbox    8291         main      20
7 X api-ssl   8729  none        main      20
```

Figure 6 Disabled services on gateway

Also disabled bandwidth-server on 2000 port:

```
[admin@MikroTik] > tool bandwidth-server print
enabled: yes
authenticate: yes
allocate-udp-ports-from: 2000
max-sessions: 100
[admin@MikroTik] > tool bandwidth-server set enabled=no
[admin@MikroTik] > tool bandwidth-server print
enabled: no
authenticate: yes
allocate-udp-ports-from: 2000
max-sessions: 100
```

Figure 7 Disable bandwidth-server on gateway

Scanning gateway again, only necessary services are available:

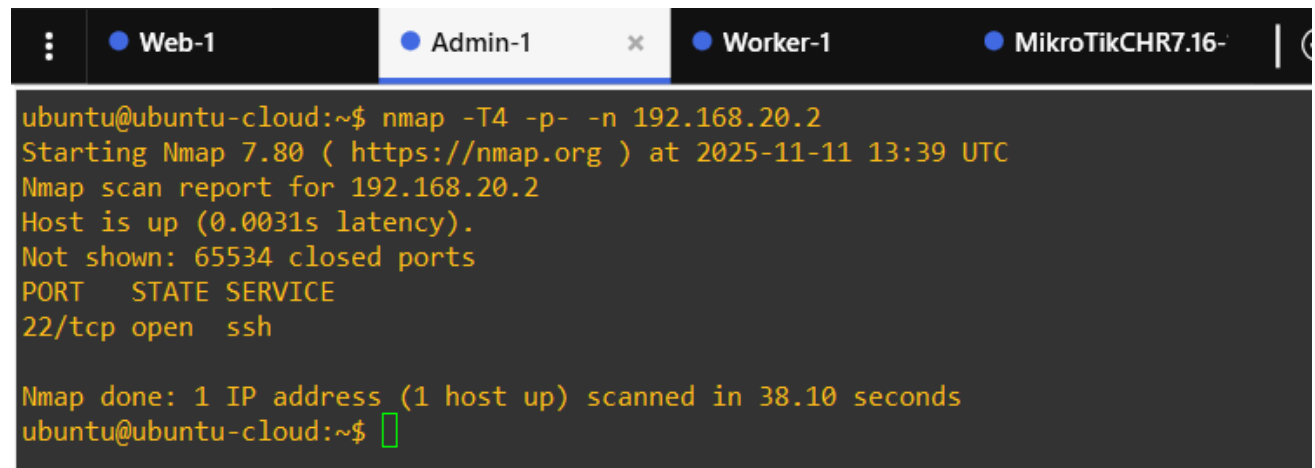
```
salekh@salekhmamadaliev25sne01:~$ nmap -T4 -p- -n 192.168.49.2
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-11 12:57 UTC
Nmap scan report for 192.168.49.2
Host is up (0.00088s latency).
Not shown: 65532 closed tcp ports (conn-refused)
PORT      STATE SERVICE
2222/tcp  open  EtherNetIP-1
2223/tcp  open  rockwell-csp2
8080/tcp  open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 26.83 seconds
```

Figure 8 Gateway scan results with disabled services

4.1 Scanning Worker from admin, in results only port 22-ssh.

Explanation of the nmap options: -T4 option speeds up scan, -p- enables scanning all ports(1-65535), -n disables DNS lookup.

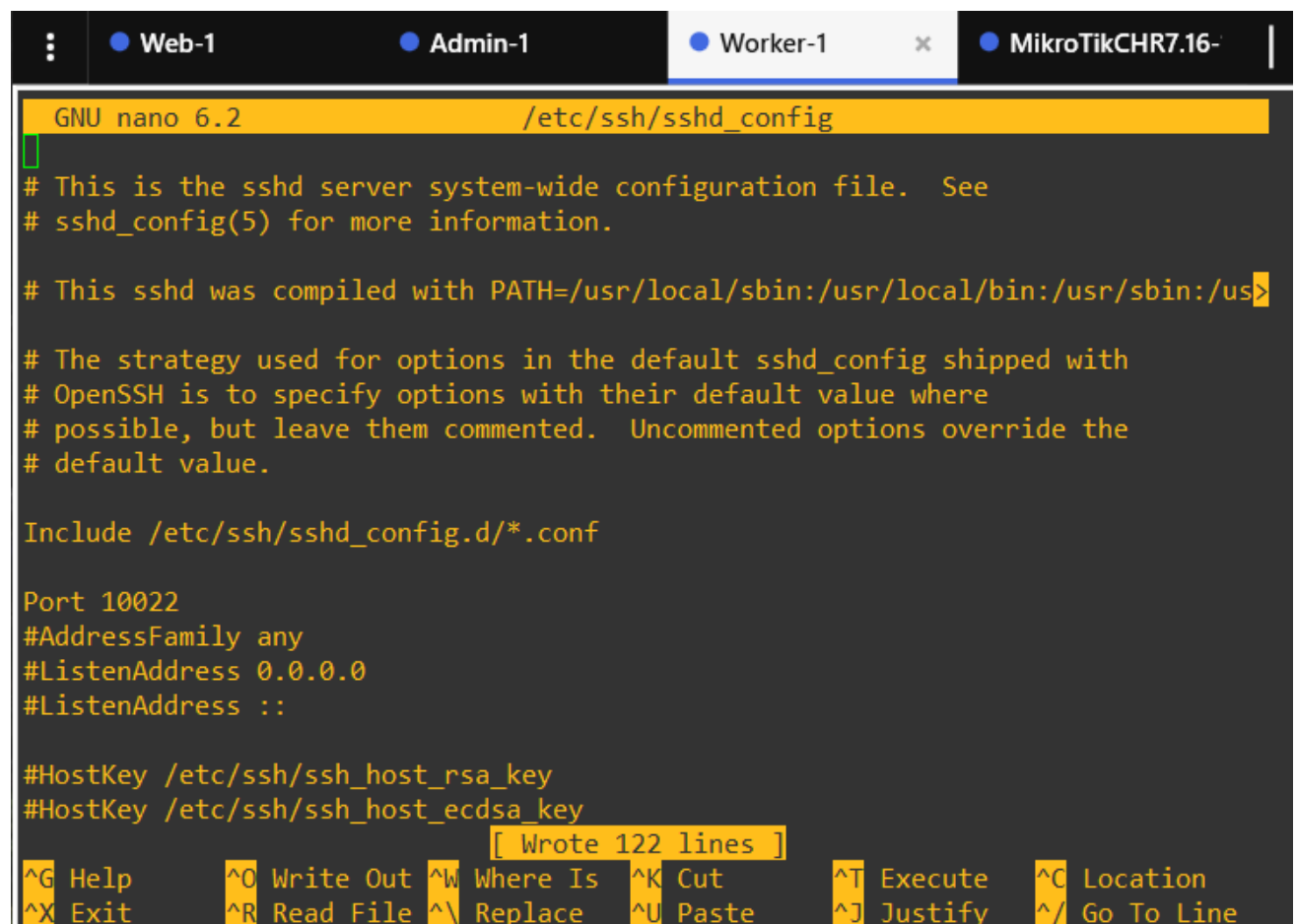


```
ubuntu@ubuntu-cloud:~$ nmap -T4 -p- -n 192.168.20.2
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-11 13:39 UTC
Nmap scan report for 192.168.20.2
Host is up (0.0031s latency).
Not shown: 65534 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 38.10 seconds
ubuntu@ubuntu-cloud:~$
```

Figure 9 Worker scan results

4.2 Changing SSH port from 22 to 10022:



```
GNU nano 6.2 /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 10022
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key

[ Wrote 122 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Figure 10 Changing SSH config

```
GNU nano 6.2 /etc/systemd/system/ssh.socket.d/listen.conf
[Socket]
ListenStream=
ListenStream=10022
```

Figure 11 Changing SSH socket config

Restarting services to apply changes and checking results:

```
ubuntu@ubuntu-cloud:~$ sudo systemctl daemon-reload
ubuntu@ubuntu-cloud:~$ sudo systemctl restart ssh.socket
ubuntu@ubuntu-cloud:~$ sudo systemctl restart ssh.service
ubuntu@ubuntu-cloud:~$ sudo ss -tulpn | grep ssh
tcp  LISTEN  0      128          0.0.0.0:10022      0.0.0.0:*    users:(("sshd",pid=1345,fd=3))
tcp  LISTEN  0      128          [::]:10022        [::]:*       users:(("sshd",pid=1345,fd=4))
ubuntu@ubuntu-cloud:~$
```

Blocking input ICMP traffic on Worker with iptables:

```
ubuntu@ubuntu-cloud:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target    prot opt source                destination
DROP      icmp -- anywhere           anywhere        icmp echo-request

Chain FORWARD (policy ACCEPT)
target    prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
ubuntu@ubuntu-cloud:~$
```

4.3 Scanning Worker without extra options, no open port found:

```
ubuntu@ubuntu-cloud:~$ nmap 192.168.20.2
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-11 13:55 UTC
Nmap scan report for 192.168.20.2
Host is up (0.0052s latency).
All 1000 scanned ports on 192.168.20.2 are closed

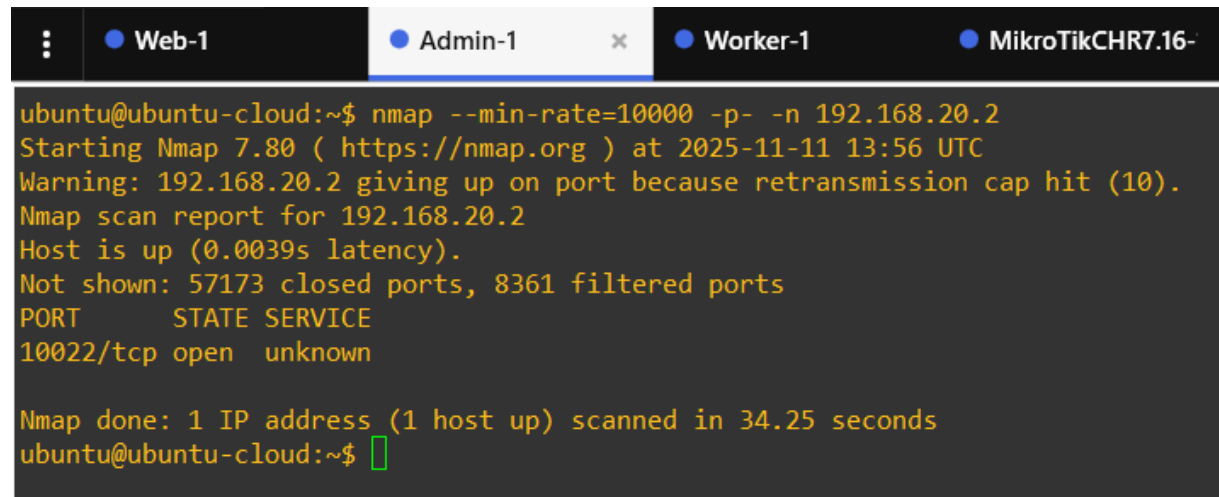
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
ubuntu@ubuntu-cloud:~$
```

Figure 12 Worker scan without arguments

4.4 Scanning Worker again with extra options:

- --min-rate=10000 significantly increases port scan;
- -p- scans all ports;
- -n disables DNS lookup.

Now open SSH port is found.

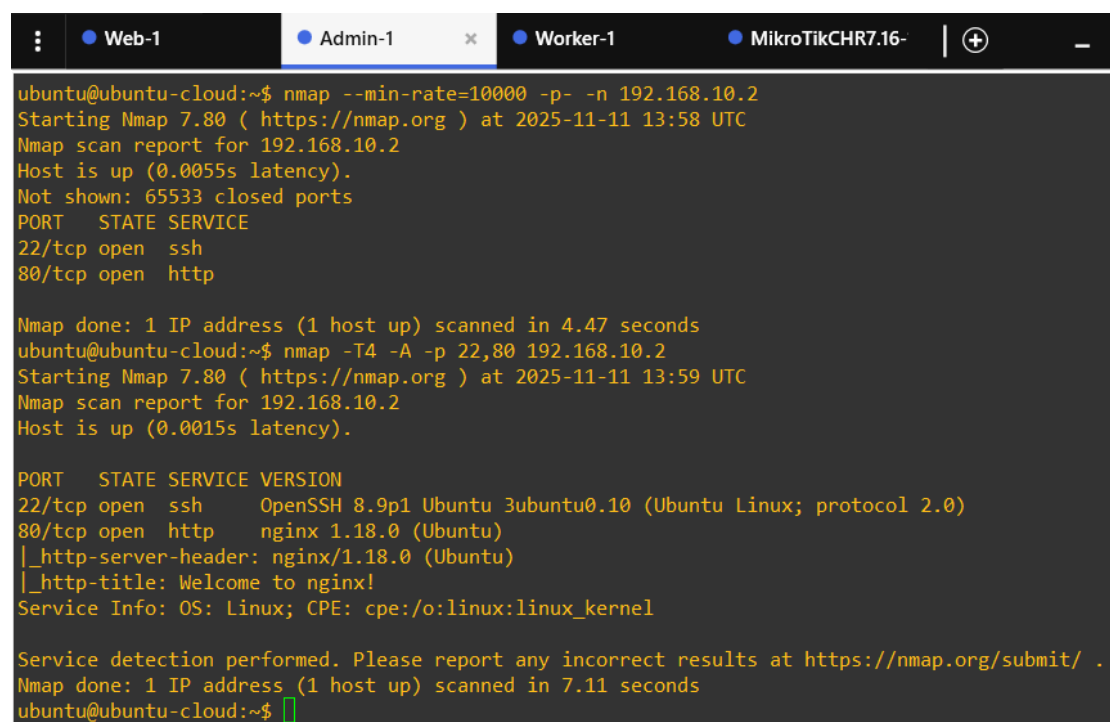


```
ubuntu@ubuntu-cloud:~$ nmap --min-rate=10000 -p- -n 192.168.20.2
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-11 13:56 UTC
Warning: 192.168.20.2 giving up on port because retransmission cap hit (10).
Nmap scan report for 192.168.20.2
Host is up (0.0039s latency).
Not shown: 57173 closed ports, 8361 filtered ports
PORT      STATE SERVICE
10022/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 34.25 seconds
ubuntu@ubuntu-cloud:~$
```

Figure 13 Worker scan with extra options

4.5 To gather more information with nmap, we can perform aggressive scan with option -A, this enables OS detection, version detection, scanning with scripts relative to the open ports, and traceroute.



```
ubuntu@ubuntu-cloud:~$ nmap --min-rate=10000 -p- -n 192.168.10.2
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-11 13:58 UTC
Nmap scan report for 192.168.10.2
Host is up (0.0055s latency).
Not shown: 65533 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 4.47 seconds
ubuntu@ubuntu-cloud:~$ nmap -T4 -A -p 22,80 192.168.10.2
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-11 13:59 UTC
Nmap scan report for 192.168.10.2
Host is up (0.0015s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.10 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     nginx 1.18.0 (Ubuntu)
|_http-server-header: nginx/1.18.0 (Ubuntu)
|_http-title: Welcome to nginx!
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.11 seconds
ubuntu@ubuntu-cloud:~$
```

Figure 14 Web aggressive scan results

Task 2 - Traffic Captures

1. Start capturing traffic on the link between Cloud and Gateway:

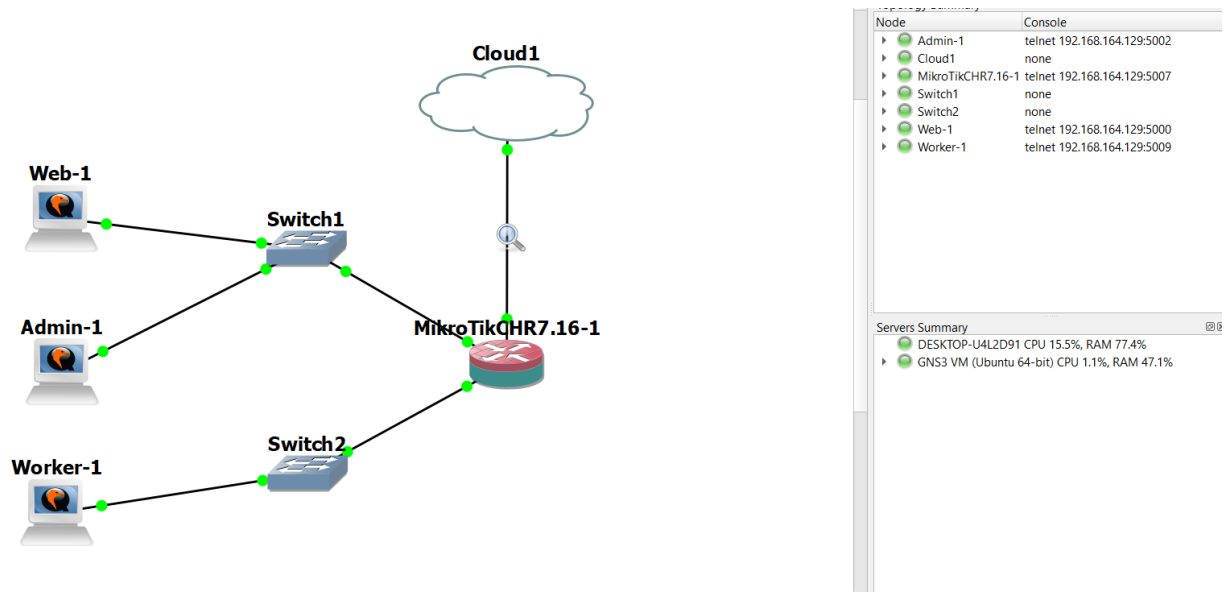


Figure 15 Capturing traffic on Cloud-Gateway link

Requesting web page from Web machine's http server from Ubuntu VM:

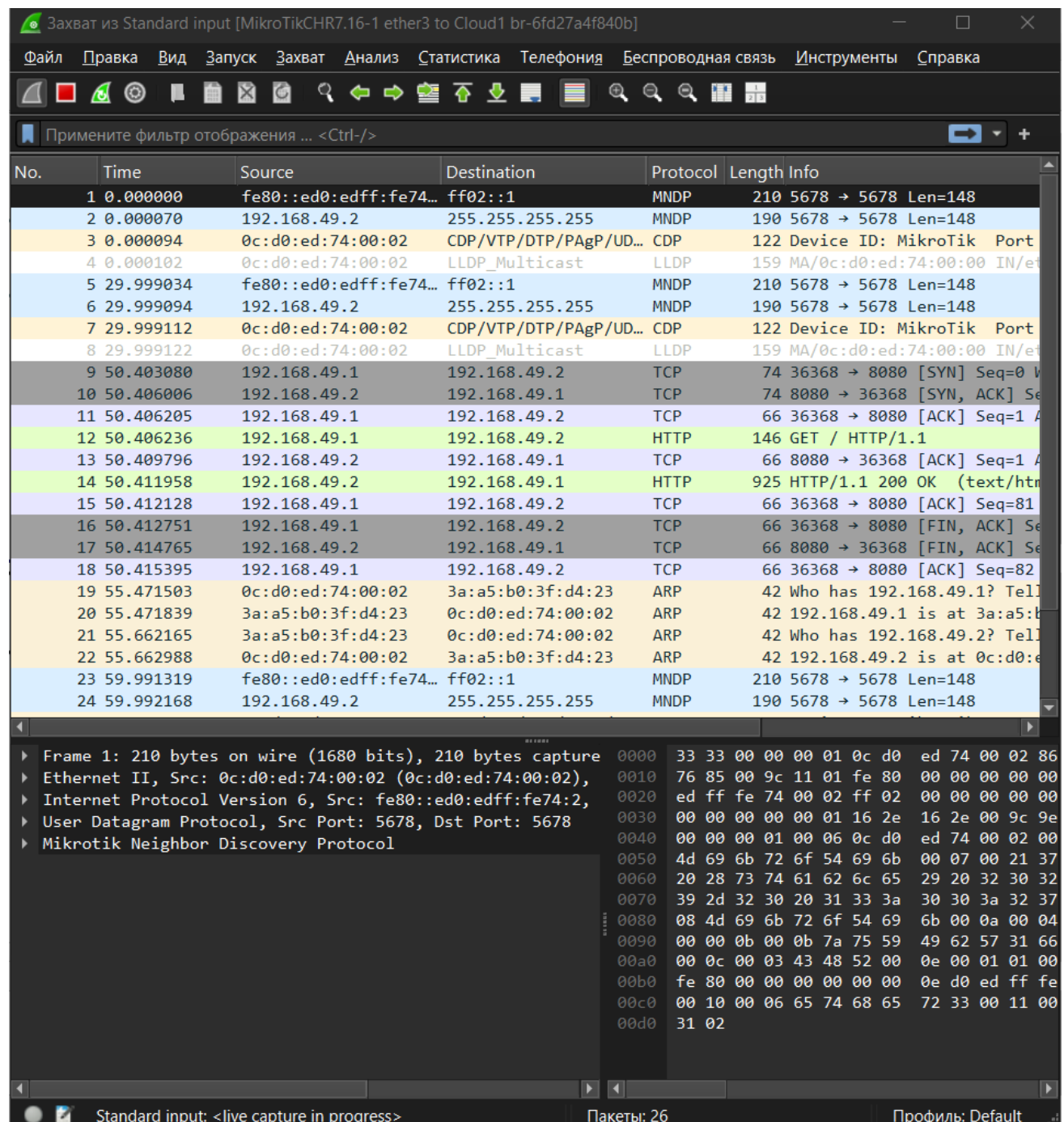
```
ubuntu_lab1.tlp - salekh@192.168.164.129:22 - Bitwise xterm - salekh@salekhmamadaliev25sne01: ~
salekh@salekhmamadaliev25sne01:~$ curl http://192.168.49.2:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
salekh@salekhmamadaliev25sne01:~$
```

Figure 16 Curl request from Ubuntu VM

Exploring captured packets via Wireshark:



The image shows the Wireshark network protocol analyzer interface. The top menu bar includes options like 'Файл', 'Правка', 'Вид', 'Запуск', 'Захват', 'Анализ', 'Статистика', 'Телефония', 'Беспроводная связь', 'Инструменты', and 'Справка'. Below the menu is a toolbar with various icons for file operations, capture, and analysis. A filter bar at the top of the packet list shows 'Примените фильтр отображения ... <Ctrl-/>'. The packet list table has columns for 'No.', 'Time', 'Source', 'Destination', 'Protocol', 'Length', and 'Info'. It displays 24 packets, including MNDP, CDP, LLDP, TCP, and HTTP. The bottom pane shows the detailed view of the selected packet (Frame 1), which is an Ethernet II frame containing an Internet Protocol Version 6 packet, which in turn contains a User Datagram Protocol packet and a Mikrotik Neighbor Discovery Protocol packet. The packet bytes are displayed in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::ed0:edff:fe74...	ff02::1	MNDP	210	5678 → 5678 Len=148
2	0.000070	192.168.49.2	255.255.255.255	MNDP	190	5678 → 5678 Len=148
3	0.000094	0c:d0:ed:74:00:02	CDP/VTP/DTP/PAGP/UD...	CDP	122	Device ID: MikroTik Port
4	0.000102	0c:d0:ed:74:00:02	LLDP_Multicast	LLDP	159	MA/0c:d0:ed:74:00:00 IN/et
5	29.999034	fe80::ed0:edff:fe74...	ff02::1	MNDP	210	5678 → 5678 Len=148
6	29.999094	192.168.49.2	255.255.255.255	MNDP	190	5678 → 5678 Len=148
7	29.999112	0c:d0:ed:74:00:02	CDP/VTP/DTP/PAGP/UD...	CDP	122	Device ID: MikroTik Port
8	29.999122	0c:d0:ed:74:00:02	LLDP_Multicast	LLDP	159	MA/0c:d0:ed:74:00:00 IN/et
9	50.403080	192.168.49.1	192.168.49.2	TCP	74	36368 → 8080 [SYN] Seq=0
10	50.406006	192.168.49.2	192.168.49.1	TCP	74	8080 → 36368 [SYN, ACK] Se
11	50.406205	192.168.49.1	192.168.49.2	TCP	66	36368 → 8080 [ACK] Seq=1
12	50.406236	192.168.49.1	192.168.49.2	HTTP	146	GET / HTTP/1.1
13	50.409796	192.168.49.2	192.168.49.1	TCP	66	8080 → 36368 [ACK] Seq=1
14	50.411958	192.168.49.2	192.168.49.1	HTTP	925	HTTP/1.1 200 OK (text/htm
15	50.412128	192.168.49.1	192.168.49.2	TCP	66	36368 → 8080 [ACK] Seq=81
16	50.412751	192.168.49.1	192.168.49.2	TCP	66	36368 → 8080 [FIN, ACK] Se
17	50.414765	192.168.49.2	192.168.49.1	TCP	66	8080 → 36368 [FIN, ACK] Se
18	50.415395	192.168.49.1	192.168.49.2	TCP	66	36368 → 8080 [ACK] Seq=82
19	55.471503	0c:d0:ed:74:00:02	3a:a5:b0:3f:d4:23	ARP	42	Who has 192.168.49.1? Tel
20	55.471839	3a:a5:b0:3f:d4:23	0c:d0:ed:74:00:02	ARP	42	192.168.49.1 is at 3a:a5:b
21	55.662165	3a:a5:b0:3f:d4:23	0c:d0:ed:74:00:02	ARP	42	Who has 192.168.49.2? Tel
22	55.662988	0c:d0:ed:74:00:02	3a:a5:b0:3f:d4:23	ARP	42	192.168.49.2 is at 0c:d0:e
23	59.991319	fe80::ed0:edff:fe74...	ff02::1	MNDP	210	5678 → 5678 Len=148
24	59.992168	192.168.49.2	255.255.255.255	MNDP	190	5678 → 5678 Len=148

Frame 1: 210 bytes on wire (1680 bits), 210 bytes capture
Ethernet II, Src: 0c:d0:ed:74:00:02 (0c:d0:ed:74:00:02),
Internet Protocol Version 6, Src: fe80::ed0:edff:fe74:2,
User Datagram Protocol, Src Port: 5678, Dst Port: 5678
Mikrotik Neighbor Discovery Protocol

Figure 17 Exploring captured traffic via Wireshark

We can see different packets in this capture: MNDP, CDP, LLDP, TCP, HTTP, ARP. But we are interested in TCP and HTTP protocols packets. On the picture above we can see TCP 3-way handshake: SYN, SYN-ACK, ACK. Then HTTP request from Ubuntu, ACK TCP response and HTTP response from Web server, and ACK from Ubuntu as well. Next three TCP packets show us that the

connection has been closed with FIN TCP packets. We can choose any TCP/HTTP packet and follow TCP or HTTP flow to see whole HTTP communication in plaintext:

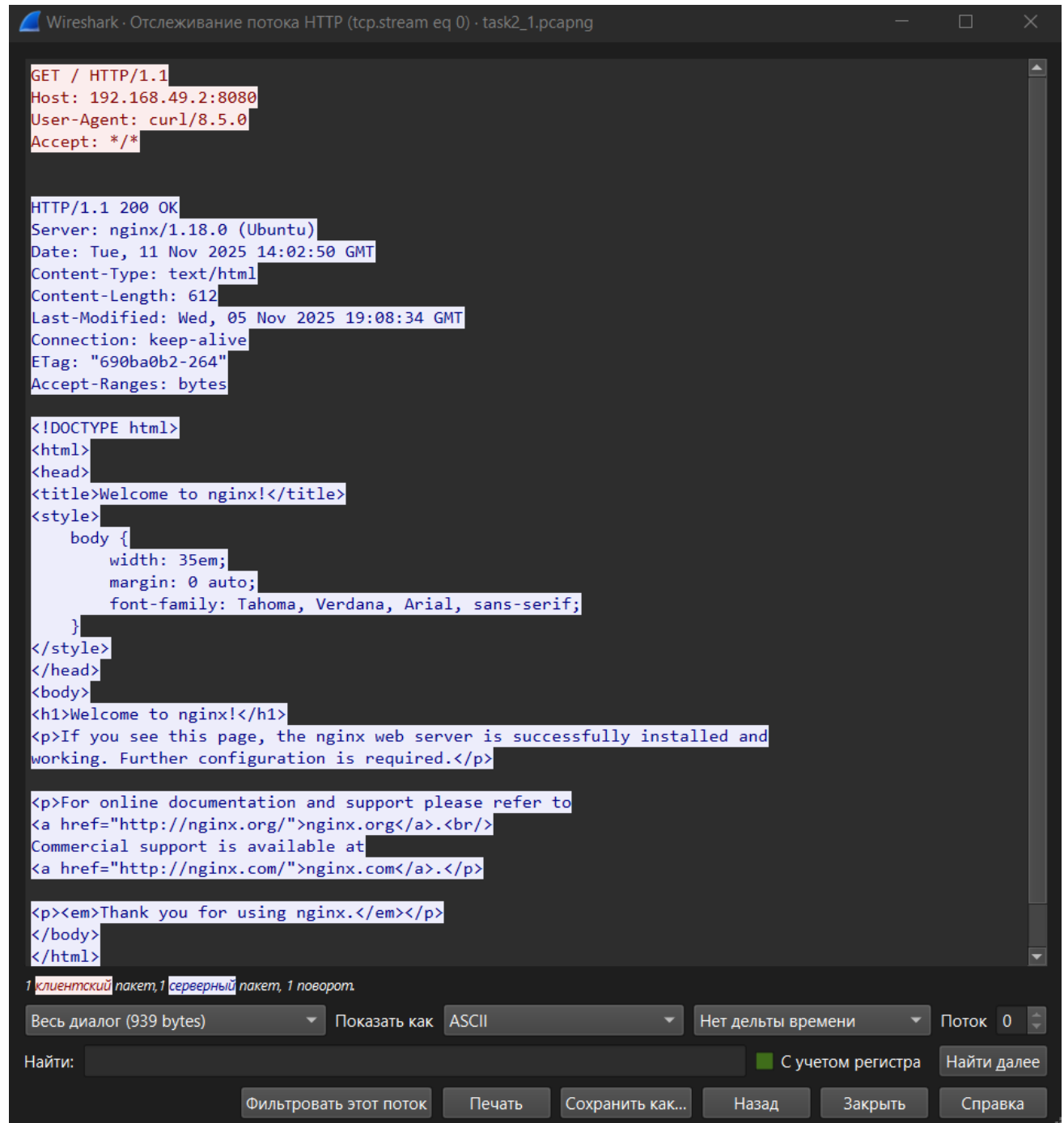
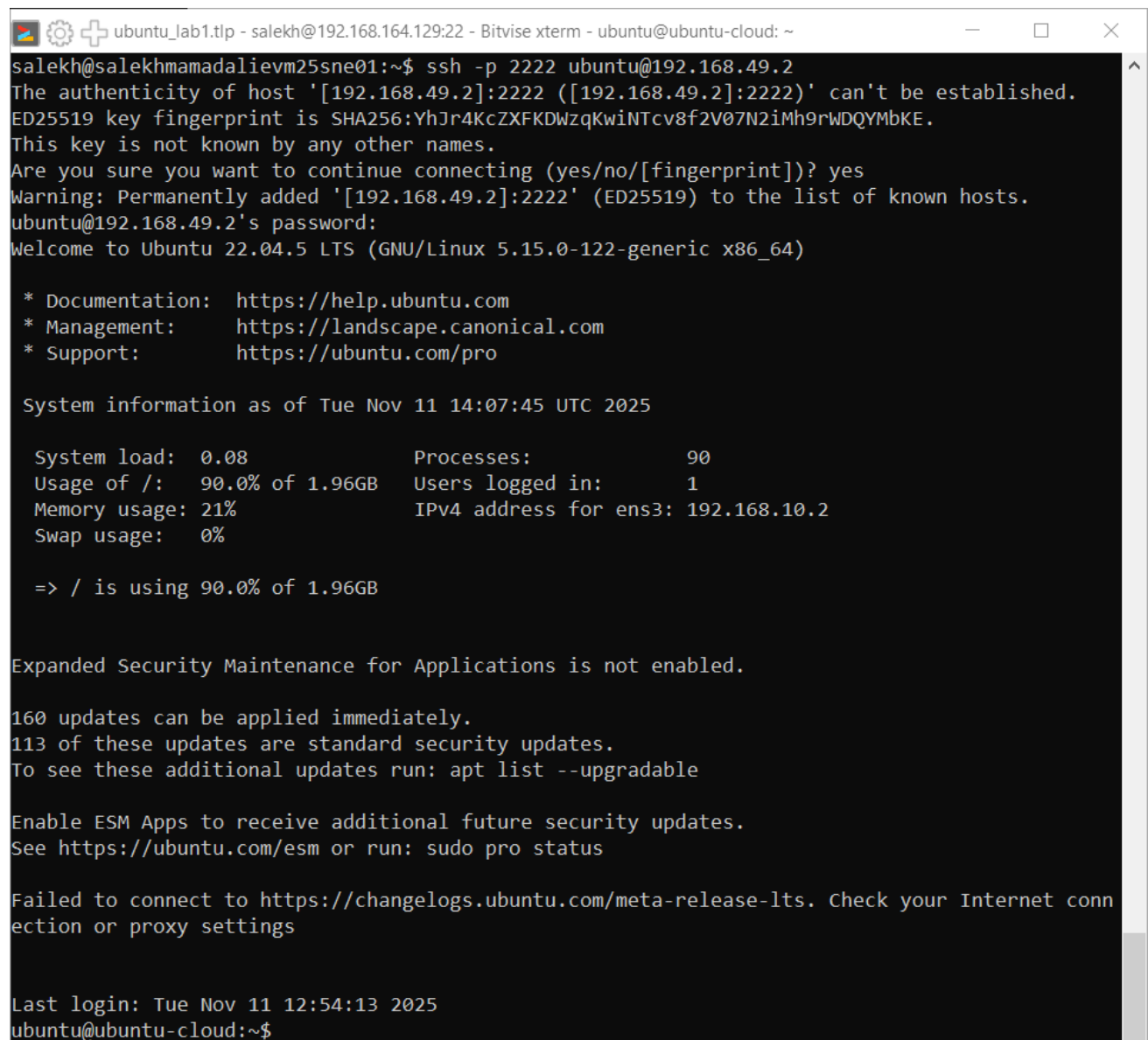


Figure 18 HTTP communication in plaintext

We can see this because there's no encryption and all HTTP requests/response can be seen to potential attacker.

HTTP headers are metadata fields in request/response, which provide additional information to server about client, and to client about server. They may have different purposes like storing client session information, define content type of message, or tell server which program is used by client to access server (user-agent header).

2. Connecting to Admin from Ubuntu VM via ssh:



```
ubuntu_lab1.tlp - salekh@192.168.164.129:22 - Bitwise xterm - ubuntu@ubuntu-cloud: ~
salekh@salekhmamadalievm25sne01:~$ ssh -p 2222 ubuntu@192.168.49.2
The authenticity of host '[192.168.49.2]:2222 ([192.168.49.2]:2222)' can't be established.
ED25519 key fingerprint is SHA256:YhJr4KcZXFkDWzqKwiNTcv8f2V07N2iMh9rWDQYmbKE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.49.2]:2222' (ED25519) to the list of known hosts.
ubuntu@192.168.49.2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-122-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Nov 11 14:07:45 UTC 2025

System load:  0.08               Processes:            90
Usage of /:   90.0% of 1.96GB    Users logged in:     1
Memory usage: 21%               IPv4 address for ens3: 192.168.10.2
Swap usage:   0%

=> / is using 90.0% of 1.96GB

Expanded Security Maintenance for Applications is not enabled.

160 updates can be applied immediately.
113 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Tue Nov 11 12:54:13 2025
ubuntu@ubuntu-cloud:~$
```

Figure 19 Connecting to Admin via SSH

This time in the capture we have mostly TCP packets.

*Standard input [MikroTikCHR7.16-1 ether3 to Cloud1 br-6fd27a4f840b]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::ed0:edff:fe74...	ff02::1	MNDP	210	5678 → 5678 Len=148
2	0.000095	192.168.49.2	255.255.255.255	MNDP	190	5678 → 5678 Len=148
3	0.000122	0c:d0:ed:74:00:02	CDP/VTP/DTP/PAGP/UD...	CDP	122	Device ID: MikroTik
4	0.000130	0c:d0:ed:74:00:02	LLDP_Multicast	LLDP	159	MA/0c:d0:ed:74:00:00
5	11.205522	192.168.49.1	192.168.49.2	TCP	74	45620 → 2222 [SYN] Se
6	11.208250	192.168.49.2	192.168.49.1	TCP	74	2222 → 45620 [SYN, AC
7	11.208435	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
8	11.208943	192.168.49.1	192.168.49.2	TCP	109	45620 → 2222 [PSH, AC
9	11.210569	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se
10	11.258520	192.168.49.2	192.168.49.1	TCP	108	2222 → 45620 [PSH, AC
11	11.258596	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
12	11.259736	192.168.49.1	192.168.49.2	TCP	1514	45620 → 2222 [ACK] Se
13	11.259806	192.168.49.1	192.168.49.2	TCP	154	45620 → 2222 [PSH, AC
14	11.261130	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se
15	11.261263	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se
16	11.264759	192.168.49.2	192.168.49.1	TCP	1178	2222 → 45620 [PSH, AC
17	11.305959	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
18	11.364545	192.168.49.1	192.168.49.2	TCP	1274	45620 → 2222 [PSH, AC
19	11.387495	192.168.49.2	192.168.49.1	TCP	1514	2222 → 45620 [ACK] Se
20	11.387582	192.168.49.2	192.168.49.1	TCP	182	2222 → 45620 [PSH, AC
21	11.388312	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
22	11.388372	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
23	12.690901	192.168.49.1	192.168.49.2	TCP	82	45620 → 2222 [PSH, AC
24	12.734268	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se

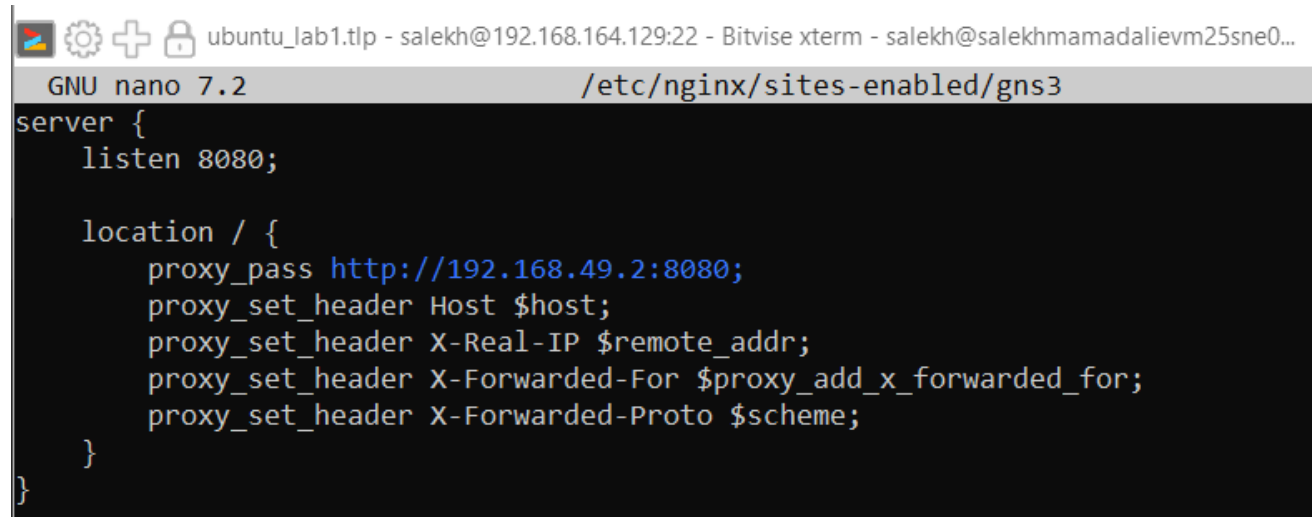
Frame 1: 210 bytes on wire (1680 bits), 210 bytes capture
 Ethernet II, Src: 0c:d0:ed:74:00:02 (0c:d0:ed:74:00:02),
 Internet Protocol Version 6, Src: fe80::ed0:edff:fe74:2,
 User Datagram Protocol, Src Port: 5678, Dst Port: 5678
 Mikrotik Neighbor Discovery Protocol

wireshark_Standard input2E15F3.pcapng Пакеты: 69 · Потеряно: 0 (0.0%) Профиль: Default

Figure 20 Fragment of captured SSH traffic

TCP flow again starting with 3-way handshake then next packets contain information about SSH version and encryption algorithms. Next packets are encrypted with negotiated algorithm and we cannot see any plaintext data.

3. First, I configured Nginx proxy server on Ubuntu VM to listen at 8080 port and proxy requests to gateway address port 8080. What is because gns3server is installed on Ubuntu VM and GNS user interface is installed on Windows host machine as well as Burp Suite:

A screenshot of a terminal window. The title bar shows icons for a file manager, settings, a plus sign, and a lock, followed by the text 'ubuntu_lab1.tlp - salekh@192.168.164.129:22 - Bitwise xterm - salekh@salekhmamadalievm25sne0...'. The terminal itself has a grey header bar that says 'GNU nano 7.2' on the left and '/etc/nginx/sites-enabled/gns3' on the right. The main area is black with white text showing the Nginx configuration for 'gns3'. The configuration starts with 'server {' followed by 'listen 8080;'. Then there is a 'location / {' block containing 'proxy_pass http://192.168.49.2:8080;', 'proxy_set_header Host \$host;', 'proxy_set_header X-Real-IP \$remote_addr;', 'proxy_set_header X-Forwarded-For \$proxy_add_x_forwarded_for;', and 'proxy_set_header X-Forwarded-Proto \$scheme;'. The block ends with '}' and the main server block ends with '}'.

```
server {  
    listen 8080;  
  
    location / {  
        proxy_pass http://192.168.49.2:8080;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
    }  
}
```

Figure 22 Nginx proxy configuration on Ubuntu VM

Next we connect to Web machine's HTTP server using internal Burp browser, which allow us to intercept HTTP requests in Burp Suite. We intercept simple GET request and can see HTTP request in burp's interface.:

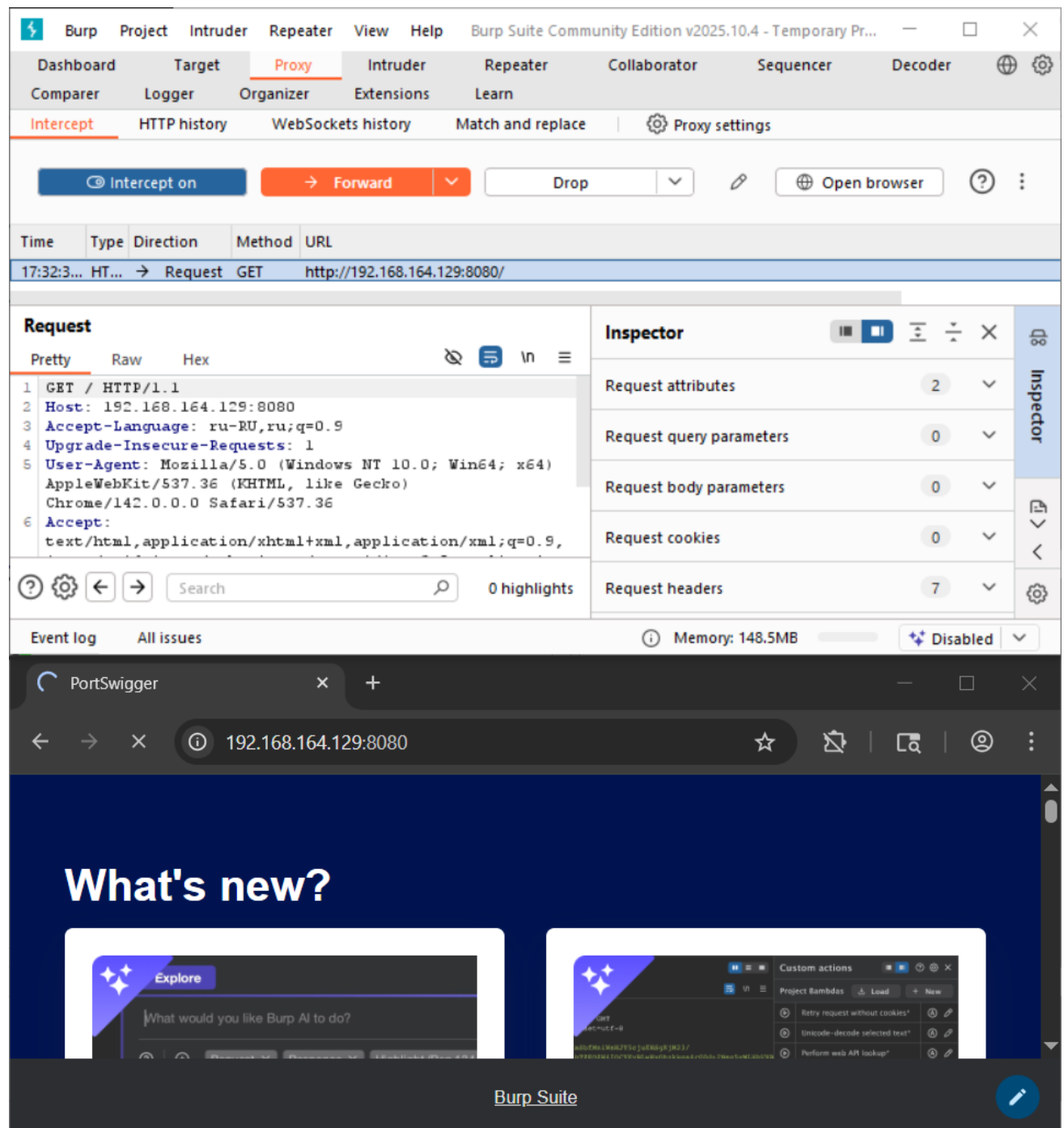


Figure 23 Intercepting HTTP request with Burp Suite

We can modify this HTTP request, for example changing the browser's user-agent to "New_user_agent". Then we can forward this request and check the results in captured traffic via Wireshark:

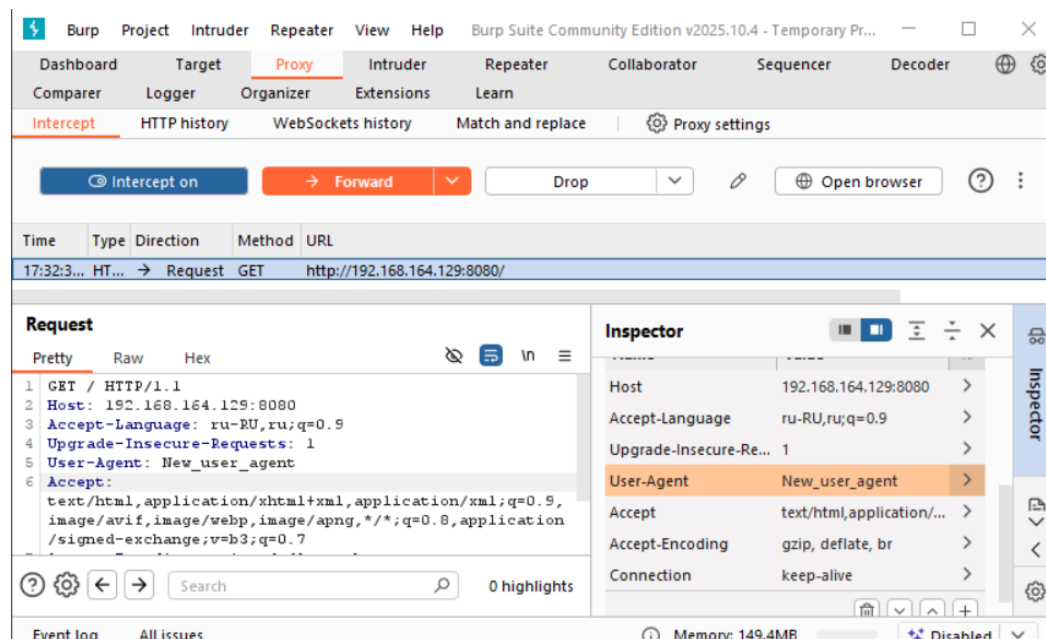


Figure 24 Modifying HTTP user-agent header

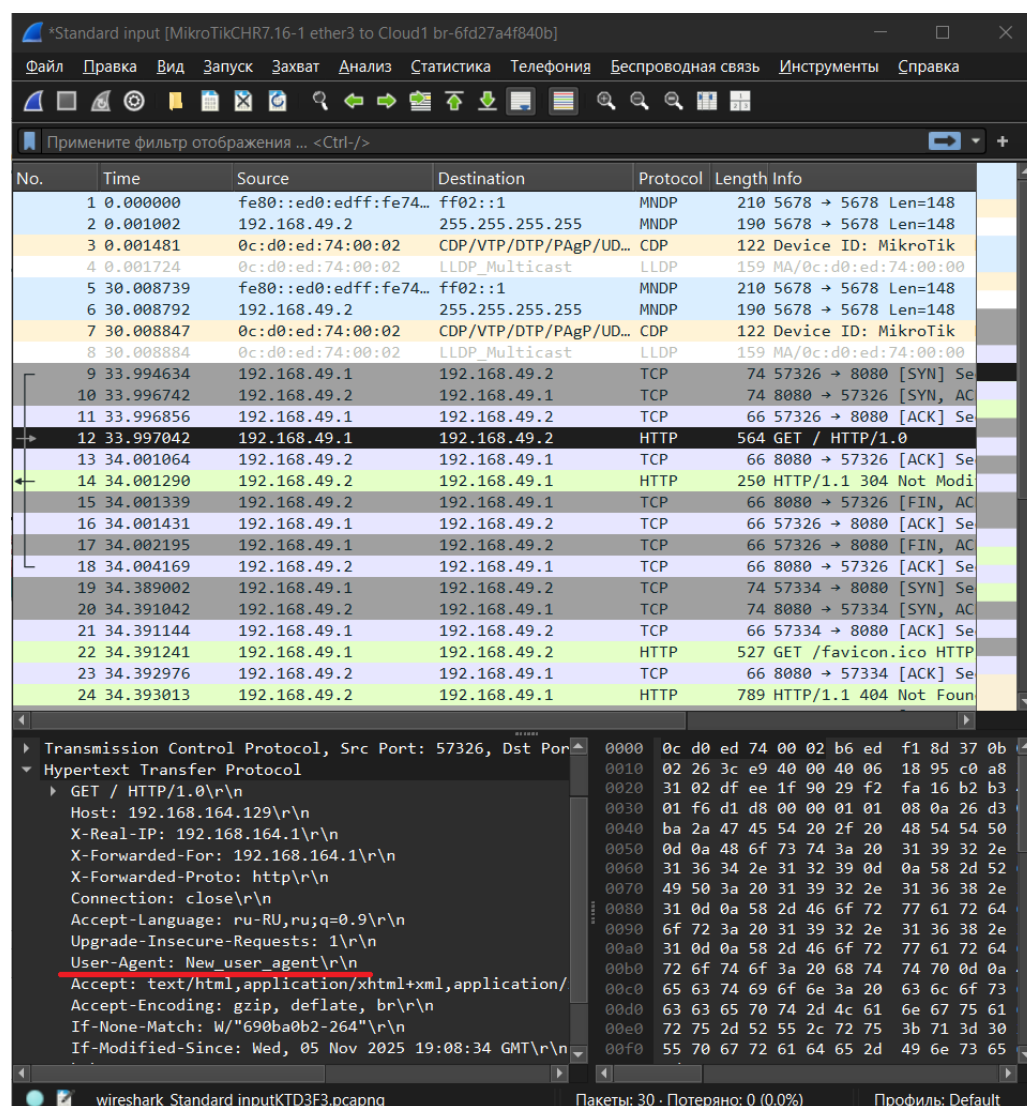
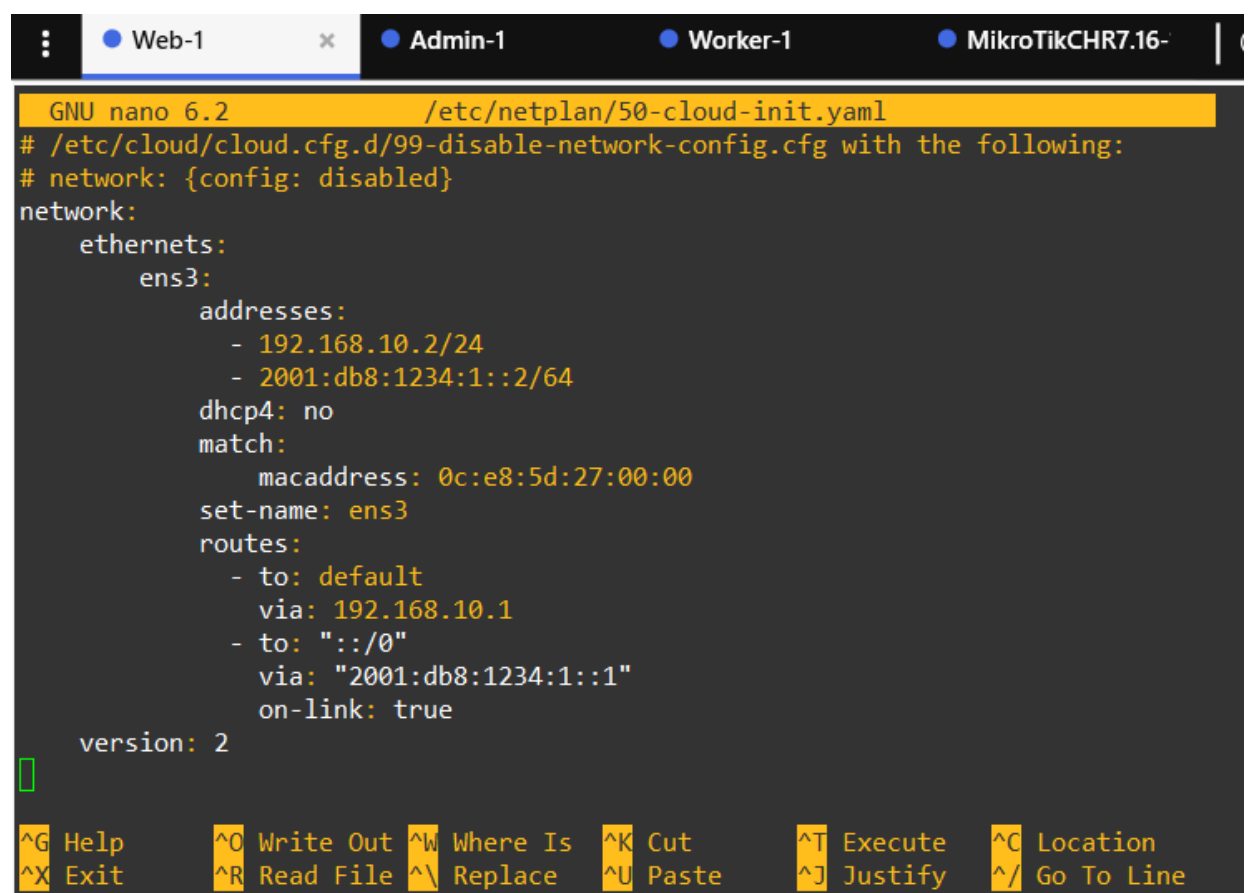


Figure 25 Captured altered HTTP packet

We are able to modify HTTP requests and not SSH requests because there's not encryption in HTTP, so requests/response are transmitted as plaintext. We can use SSL/TLS protocol with HTTP to add secrecy and prevent possible man in the middle attacks.

Task 3 - IPv6

1. Configured IPv6 addresses on each machine:



```
GNU nano 6.2 /etc/netplan/50-cloud-init.yaml
# /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg with the following:
# network: {config: disabled}
network:
  ethernets:
    ens3:
      addresses:
        - 192.168.10.2/24
        - 2001:db8:1234:1::2/64
      dhcp4: no
      match:
        macaddress: 0c:e8:5d:27:00:00
      set-name: ens3
      routes:
        - to: default
          via: 192.168.10.1
        - to: "::/0"
          via: "2001:db8:1234:1::1"
          on-link: true
      version: 2
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

Figure 26 Netplan configuration on Web

```
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:e8:5d:27:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.10.2/24 brd 192.168.10.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 2001:db8:1234:1::2/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::ee8:5dff:fe27:0/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$ ip -6 r
::1 dev lo proto kernel metric 256 pref medium
2001:db8:1234:1::/64 dev ens3 proto kernel metric 256 pref medium
fe80::/64 dev ens3 proto kernel metric 256 pref medium
default via 2001:db8:1234:1::1 dev ens3 proto static metric 1024 onlink pref medium
ubuntu@ubuntu-cloud:~$
```

Figure 27 Applied configuration on Web

```
GNU nano 6.2 /etc/netplan/50-cloud-init.yaml
# /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg with the following:
# network: {config: disabled}
network:
  ethernets:
    ens3:
      addresses:
        - 192.168.10.3/24
        - 2001:db8:1234:1::3/64
      dhcp4: no
      match:
        macaddress: 0c:03:a9:6b:00:00
      set-name: ens3
      routes:
        - to: default
          via: 192.168.10.1
        - to: "::/0"
          via: "2001:db8:1234:1::1"
          on-link: true
      version: 2

```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line

Figure 28 Netplan configuration on Admin

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:03:a9:6b:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.10.3/24 brd 192.168.10.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 2001:db8:1234:1::3/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::e03:a9ff:fe6b:0/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$ ip r
default via 192.168.10.1 dev ens3 proto static
192.168.10.0/24 dev ens3 proto kernel scope link src 192.168.10.3
ubuntu@ubuntu-cloud:~$ ip -6 r
::1 dev lo proto kernel metric 256 pref medium
2001:db8:1234:1::/64 dev ens3 proto kernel metric 256 pref medium
fe80::/64 dev ens3 proto kernel metric 256 pref medium
default via 2001:db8:1234:1::1 dev ens3 proto static metric 1024 onlink pref medium
ubuntu@ubuntu-cloud:~$ 

```

Figure 29 Applied configuration on Admin

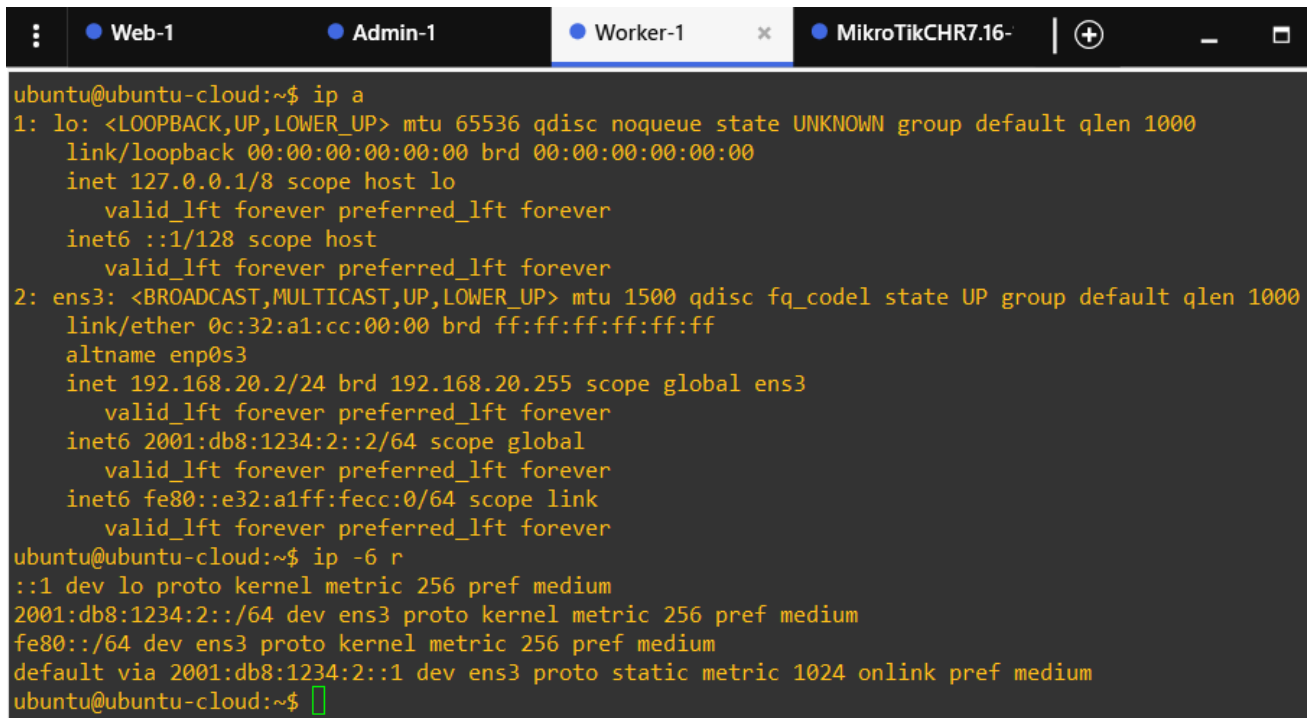
```

GNU nano 6.2 /etc/netplan/50-cloud-init.yaml
# /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg with the following:
# network: {config: disabled}
network:
  ethernets:
    ens3:
      addresses:
        - 192.168.20.2/24
        - 2001:db8:1234:2::2/64
      dhcp4: true
      match:
        macaddress: 0c:32:a1:cc:00:00
      set-name: ens3
      routes:
        - to: default
          via: 192.168.20.1
        - to: "::/0"
          via: "2001:db8:1234:2::1"
          on-link: true
  version: 2

```

[^]G Help [^]O Write Out [^]W Where Is [^]K Cut [^]T Execute [^]C Location
[^]X Exit [^]R Read File [^]\ Replace [^]U Paste [^]J Justify [^]/ Go To Line

Figure 30 Netplan configuration on Worker



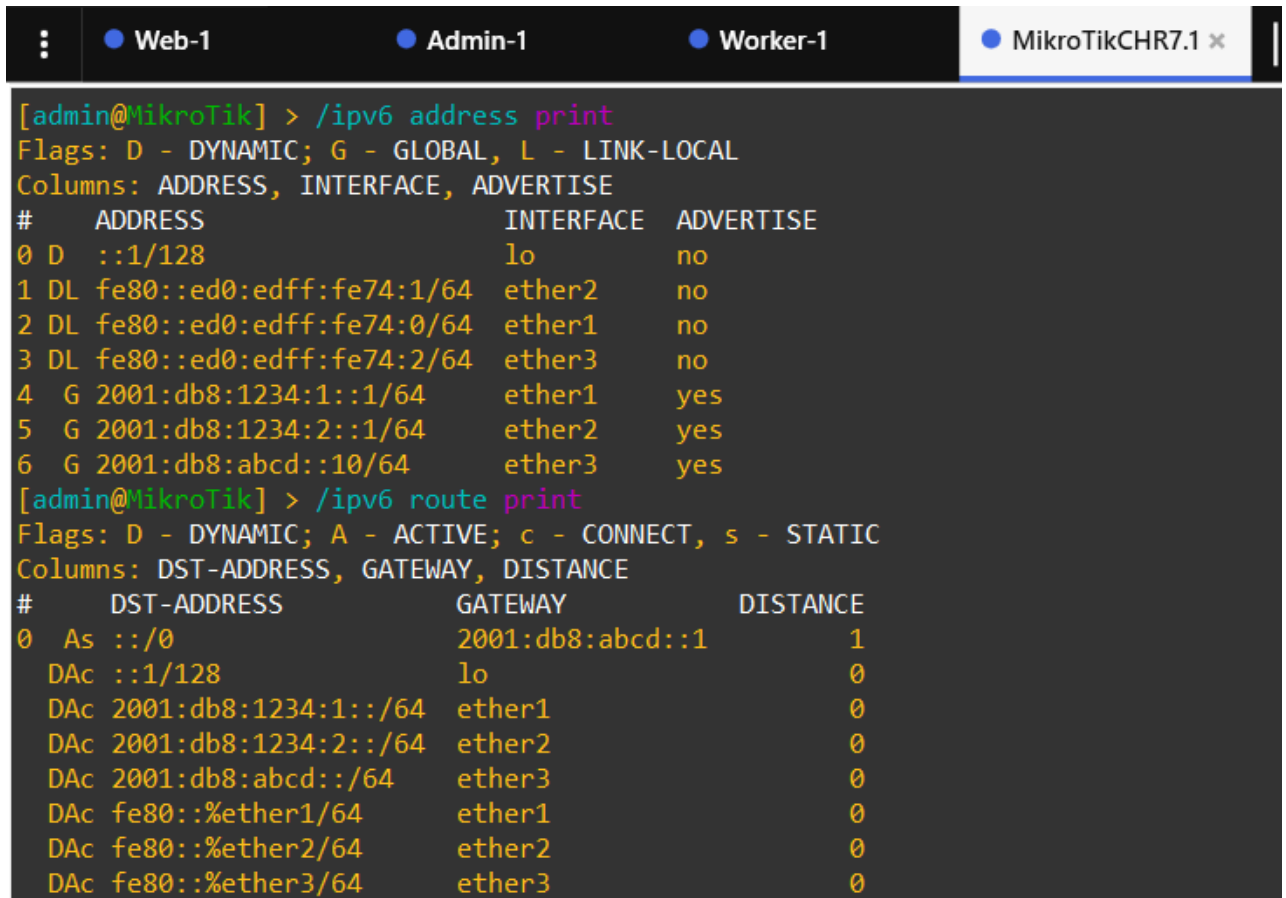
The terminal window shows the output of the 'ip a' and 'ip -6 r' commands on a Worker-1 node. The 'ip a' command shows details for the loopback interface 'lo' and the ethernet interface 'ens3'. The 'ip -6 r' command shows the IPv6 routing table.

```

ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:32:a1:cc:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.20.2/24 brd 192.168.20.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 2001:db8:1234:2::2/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::e32:a1ff:fecc:0/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$ ip -6 r
::1 dev lo proto kernel metric 256 pref medium
2001:db8:1234:2::/64 dev ens3 proto kernel metric 256 pref medium
fe80::/64 dev ens3 proto kernel metric 256 pref medium
default via 2001:db8:1234:2::1 dev ens3 proto static metric 1024 onlink pref medium
ubuntu@ubuntu-cloud:~$

```

Figure 31 Applied configuration on Worker



The terminal window shows the output of the '/ipv6 address print' and '/ipv6 route print' commands on a MikroTik Gateway. The first command shows the configured IPv6 addresses on the interfaces. The second command shows the configured IPv6 routes.

```

[admin@MikroTik] > /ipv6 address print
Flags: D - DYNAMIC; G - GLOBAL, L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
0 D ::1/128 lo no
1 DL fe80::ed0:edff:fe74:1/64 ether2 no
2 DL fe80::ed0:edff:fe74:0/64 ether1 no
3 DL fe80::ed0:edff:fe74:2/64 ether3 no
4 G 2001:db8:1234:1::/64 ether1 yes
5 G 2001:db8:1234:2::/64 ether2 yes
6 G 2001:db8:abcd::10/64 ether3 yes
[admin@MikroTik] > /ipv6 route print
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, s - STATIC
Columns: DST-ADDRESS, GATEWAY, DISTANCE
# DST-ADDRESS GATEWAY DISTANCE
0 As ::/0 2001:db8:abcd::1 1
Dac ::1/128 lo 0
Dac 2001:db8:1234:1::/64 ether1 0
Dac 2001:db8:1234:2::/64 ether2 0
Dac 2001:db8:abcd::/64 ether3 0
Dac fe80::%ether1/64 ether1 0
Dac fe80::%ether2/64 ether2 0
Dac fe80::%ether3/64 ether3 0

```

Figure 32 Configured addresses and routes on Gateway

Then check connectivity between machines:

```
ubuntu@ubuntu-cloud:~$ ping -6 2001:db8:1234:1::2
PING 2001:db8:1234:1::2(2001:db8:1234:1::2) 56 data bytes
64 bytes from 2001:db8:1234:1::2: icmp_seq=1 ttl=63 time=8.30 ms
64 bytes from 2001:db8:1234:1::2: icmp_seq=2 ttl=63 time=5.92 ms
64 bytes from 2001:db8:1234:1::2: icmp_seq=3 ttl=63 time=5.75 ms
^C
--- 2001:db8:1234:1::2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 5.747/6.658/8.303/1.165 ms
ubuntu@ubuntu-cloud:~$ ping -6 2001:db8:1234:1::3
PING 2001:db8:1234:1::3(2001:db8:1234:1::3) 56 data bytes
64 bytes from 2001:db8:1234:1::3: icmp_seq=1 ttl=63 time=8.13 ms
64 bytes from 2001:db8:1234:1::3: icmp_seq=2 ttl=63 time=7.82 ms
64 bytes from 2001:db8:1234:1::3: icmp_seq=3 ttl=63 time=6.92 ms
^C
--- 2001:db8:1234:1::3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 6.922/7.623/8.134/0.512 ms
ubuntu@ubuntu-cloud:~$
```

Figure 33 Checking IPv6 connectivity

2. Start capturing traffic on the link between Admin and Switch1:

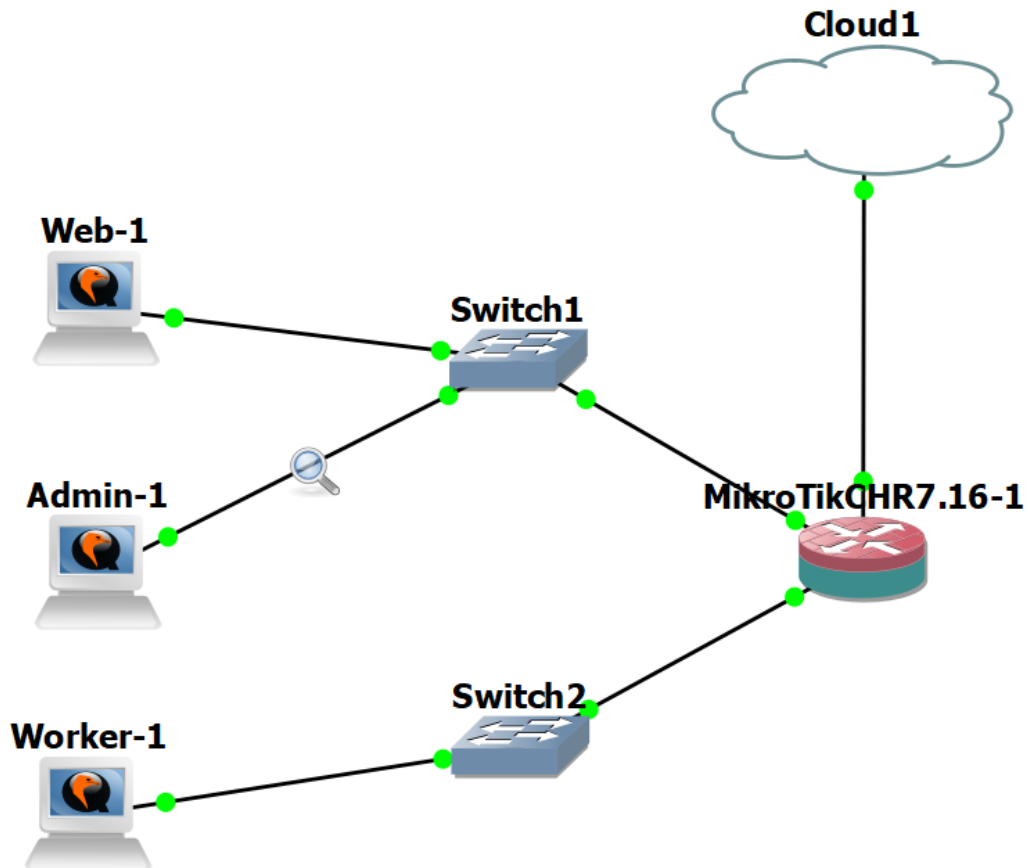


Figure 34 Capturing traffic on Admin-Switch1 link

Next accessing Web's http server using it's IPv6 address:

```
ubuntu@ubuntu-cloud:~$ curl -I http://[2001:db8:1234:1::2]/
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 12 Nov 2025 09:09:25 GMT
Content-Type: text/html
Content-Length: 612
Last-Modified: Wed, 05 Nov 2025 19:08:34 GMT
Connection: keep-alive
ETag: "690ba0b2-264"
Accept-Ranges: bytes

ubuntu@ubuntu-cloud:~$
```

Figure 35 HTTP request to Web via curl

*Standard input [Admin-1 Ethernet0 to Switch1 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
16	25.263861	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Trans
17	29.998222	fe80::ed0:edff:fe74...	ff02::1	MNDP	230	5678 → 5678 Len=168
18	29.998365	192.168.10.1	255.255.255.255	MNDP	210	5678 → 5678 Len=168
19	29.998425	0c:d0:ed:74:00:00	CDP/VTP/DTP/PAGP/UD...	CDP	122	Device ID: MikroTik
20	29.998480	0c:d0:ed:74:00:00	LLDP_Multicast	LLDP	185	MA/0c:d0:ed:74:00:00
21	31.838784	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Trans
22	33.632340	2001:db8:1234:1::3	ff02::1:ff00:2	ICMPv6	86	Neighbor Solicitation
23	33.634741	2001:db8:1234:1::2	2001:db8:1234:1::3	ICMPv6	86	Neighbor Advertisemen
24	33.636013	2001:db8:1234:1::3	2001:db8:1234:1::2	TCP	94	32906 → 80 [SYN] Seq=
25	33.637862	2001:db8:1234:1::2	2001:db8:1234:1::3	TCP	94	80 → 32906 [SYN, ACK]
26	33.640810	2001:db8:1234:1::3	2001:db8:1234:1::2	TCP	86	32906 → 80 [ACK] Seq=
27	33.643436	2001:db8:1234:1::3	2001:db8:1234:1::2	HTTP	171	HEAD / HTTP/1.1
28	33.646101	2001:db8:1234:1::2	2001:db8:1234:1::3	TCP	86	80 → 32906 [ACK] Seq=
29	33.646270	2001:db8:1234:1::2	2001:db8:1234:1::3	HTTP	333	HTTP/1.1 200 OK
30	33.649542	2001:db8:1234:1::3	2001:db8:1234:1::2	TCP	86	32906 → 80 [ACK] Seq=
31	33.728740	2001:db8:1234:1::3	2001:db8:1234:1::2	TCP	86	32906 → 80 [FIN, ACK]
32	33.730379	2001:db8:1234:1::2	2001:db8:1234:1::3	TCP	86	80 → 32906 [FIN, ACK]
33	33.731856	2001:db8:1234:1::3	2001:db8:1234:1::2	TCP	86	32906 → 80 [ACK] Seq=
34	33.781911	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Trans
35	35.803306	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Trans
36	37.146005	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Trans
37	38.748794	fe80::ee8:5dff:fe27...	2001:db8:1234:1::3	ICMPv6	86	Neighbor Solicitation
38	38.751367	2001:db8:1234:1::3	fe80::ee8:5dff:fe27...	ICMPv6	78	Neighbor Advertisemen
39	40.479729	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Trans

▶ Frame 23: 86 bytes on wire (688 bits), 86 bytes captured
 ▶ Ethernet II, Src: 0c:e8:5d:27:00:00 (0c:e8:5d:27:00:00),
 ▶ Internet Protocol Version 6, Src: 2001:db8:1234:1::2, Dst
 ▶ Internet Control Message Protocol v6
 Type: Neighbor Advertisement (136)
 Code: 0
 Checksum: 0xebc2 [correct]
 [Checksum Status: Good]
 Flags: 0x60000000, Solicited, Override
 Target Address: 2001:db8:1234:1::2
 ICMPv6 Option (Target link-layer address : 0c:e8:5d:27

0000 0c 03 a9 6b 00 00 0c e8 5d 27 00 00 86
 0010 00 00 00 20 3a ff 20 01 0d b8 12 34 00
 0020 00 00 00 00 00 02 20 01 0d b8 12 34 00
 0030 00 00 00 00 00 03 88 00 eb c2 60 00 00
 0040 0d b8 12 34 00 01 00 00 00 00 00 00 00
 0050 0c e8 5d 27 00 00

Figure 36 Captured HTTP IPv6 traffic

The main difference between IPv6 and IPv4 is with packets on Network layer. For IPv4 we have variable IP header size, from 20-60 bytes. For IPv6 we have fixed IP header size - 40 bytes.

The format of communicating remains almost the same, there's no changes in TCP or HTTP protocol layers.

We can also notice that instead of ARP requests, the ICMPv6 NDP neighbor solicitation requests are used to translate IP address to MAC address.

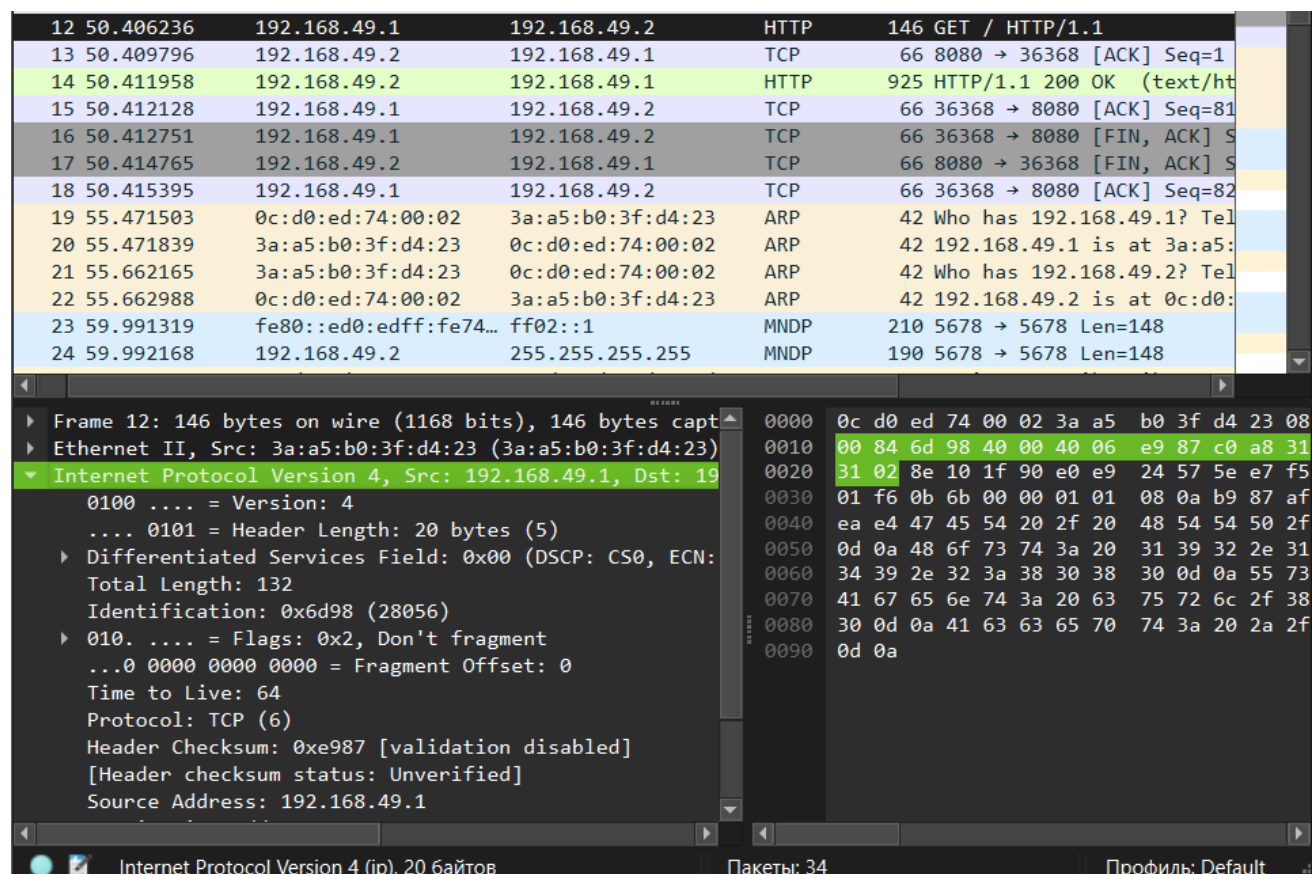


Figure 37 IP header in IPv4 HTTP request

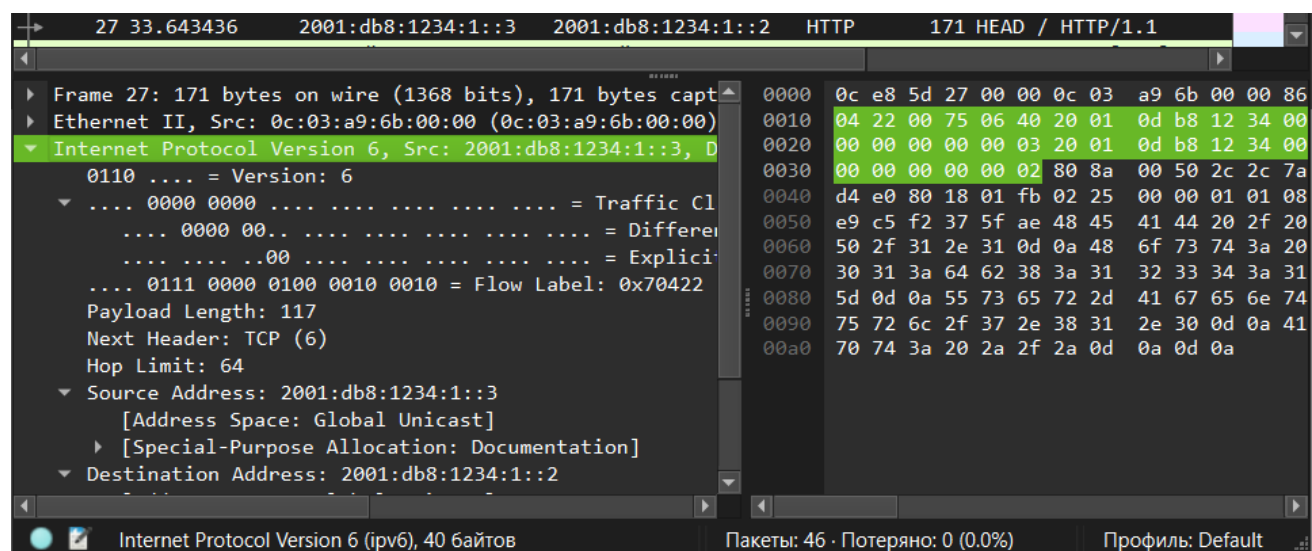


Figure 38 IP header in IPv6 HTTP request

3. For practice I will take Web and Worker IPv6 addressed:

2001:db8:1234:1::2/64 and 2001:db8:1234:2::2/64

Web's compressed address: 2001:db8:1234:1::2

First, we decompressing "::", this represents the group of consecutive zero hex digits. This can be used only once in address and we need to get 8 groups of 4 hex digits, now we have 5 groups, so we need to add three groups of zeros:

2001:db8:1234:1:0000:0000:0000:2.

Next step is to prepend zeros in each group less than 4 digits. in our example this are groups 2, 4 and 8, prepending zeros there:

2001:0db8:1234:0001:0000:0000:0000:0002.

Web's decompressed address: 2001:0db8:1234:0001:0000:0000:0000:0002

In this address octet "0001" represents subnet ID, and octet "0002" represents host ID in this subnet. /64 is subnet prefix.

Worker's compressed address: 2001:db8:1234:2::2

For Worker's address we doing completely the same steps. Decompressing chain of zeros:

2001:db8:1234:2:0000:0000:0000:2.

Then prepending zeros in each group to get 4 digits in each group:

2001:0db8:1234:0002:0000:0000:0000:0002.

Worker's decompressed address: 2001:0db8:1234:0002:0000:0000:0000:0002

References:

1. Burp Suite community edition.
<https://portswigger.net/burp/communitydownload>
2. Wireshark. <https://www.wireshark.org/docs/>
3. HTTP. <https://en.wikipedia.org/wiki/HTTP>
4. Secure Shell. https://en.wikipedia.org/wiki/Secure_Shell
5. IPv6. <https://en.wikipedia.org/wiki/IPv6>
6. Nmap. <https://nmap.org/>